# 1. Clinical Question & Core Hypothesis

## Clinical Question

The Ontologist highlights a tension where process-driven PD-L1 upregulation should enable immune escape and promote adenocarcinoma/invasive progression, yet imaging- or histology-detected immune infiltration (and early precursor states like CIS) do not always prevent progression. Why does immune presence sometimes fail to constrain invasion, and what role does early TP53 disruption play?

## Working Hypothesis

In breast cancers on a CIS-to-invasive trajectory, concurrent high tumor-infiltrating lymphocyte density with PD-L1 upregulation (CD274) and TP53 mutation-driven genomic instability enables immune evasion and early invasion, so high TIL density does not guarantee containment. Consequently, tumors with the combination of high TIL density, PD-L1 upregulation, and TP53 mutation will exhibit shorter progression-free intervals than tumors lacking this PD-L1/TP53 axis.

| Element | Value |
|---|---|
| Imaging feature | Tumor-Infiltrating Lymphocyte (TIL) density |
| Anchor gene(s) | CD274 (PD-L1), TP53 |
| Primary endpoint | PFI |

# 2. Why This Gene — Not Others

## Primary hypothesis gene(s)

**CD274 (PD-L1):** Selected as a core mechanistic driver from Tier 1 knowledge graph analysis.

**TP53:** TP53 disruption play? (2) Tier 1 paths used: I built primarily on Core Path 1 (Process → PD-L1 → adenocarcinoma) because it provides the immune-evasion mechanism, and Core Path 3 (CIS ↔ TP53 mutation) because TP53 mutations provide an early genomic scaffold for clonal progression; these two together most directly explain how immune suppression and genomic instability could interact.

## Alternative / secondary genes considered

| Tier | Gene | Role / Reason excluded from core |
|---|---|---|
| Secondary | CDH1 (E-cadherin) | Supporting modifier from knowledge graph |
| Secondary | Klebsiella (microbial inflammation) | Supporting modifier from knowledge graph |

| | | |
|---|---|---|
| Secondary | CIS (carcinoma in situ) | Supporting modifier from knowledge graph |

# 3. Clinically Testable Predictions

**Prediction 1:** TIL density (imaging-derived or transcriptomic proxies),

**Prediction 2:** TP53 mutation status, with progression-free interval (PFI) as the clinical endpoint. Validation in an independent cohort (e.g., METABRIC) strengthens inference.

**Prediction 3:** In retrospective cohort data, the prognostic benefit of the imaging feature will be modified by the anchor gene expression level (significant interaction term).

# 4. Minimum Viable Validation Plan

## A. Retrospective Cohort Analysis

**Data source:** TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

## B. Tissue-Level Validation

**Methods:** Multiplex IF / IHC / spatial transcriptomics

**Key readouts:** - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

**Goal:** Confirm that the proposed mechanism operates at the spatial / tissue level.

## C. Functional Validation

**Models:** Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

**Perturbations:** Genetic (KO/rescue) or pharmacologic modulation of anchor gene

**Readouts:** T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

**Goal:** Establish causal relationship between anchor gene activity and immune efficacy.

# 5. Strengths, Caveats & Alternative Directions

## Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

## Caveats & Risks

- Gene expression does not equal protein activity; phospho-protein or functional assays are needed for confirmation.

## Alternative Directions

- CDH1 (E-cadherin): Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- Klebsiella (microbial inflammation): Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.